

3rd-Order Active High-Pass Filter

Design for Tweeter Protection

EEE308L — Electronic Circuits II | Section: 06 | Spring 2026

BRAC University — Dept. of Electrical & Electronic Engineering

Submitted to: Saif Ahmed Sunny, Associate Professor

Group 4

Group Members

EEE308L — Design Project | Spring 2026

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Problem Statement & Technical Specifications

Sony Corporation tweeter protection — EEE308L

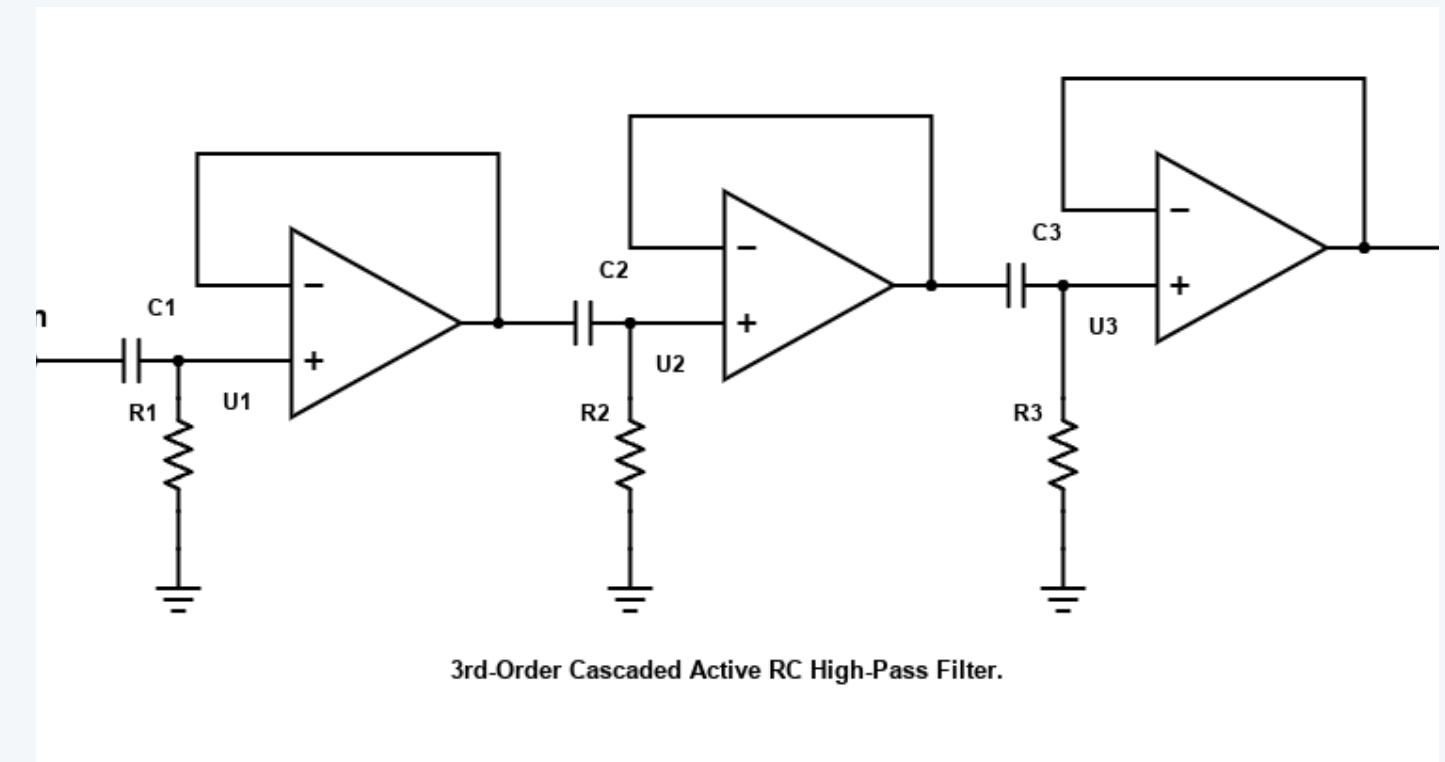
Problem Statement

- Design a 3rd-order Active High-Pass Filter
- Protects tweeter from low-frequency damage
- Passes frequencies above f_{thres}
- Rejects frequencies below f_{thres}
- Application: Sony tweeter crossover circuit

Project Requirements

- 3rd-Order Active High-Pass Filter
- Tweeter protection application
- Passes frequencies above 14 kHz
- Rejects frequencies below 14 kHz
- Roll-off: -60 dB/decade
- Uses UA741 Op-Amp
- Cascaded Active RC configuration
- $\pm 15V$ dual power supply
- Simulation in PSpice
- Hardware implementation on breadboard

Filter Circuit (Design Schematic)

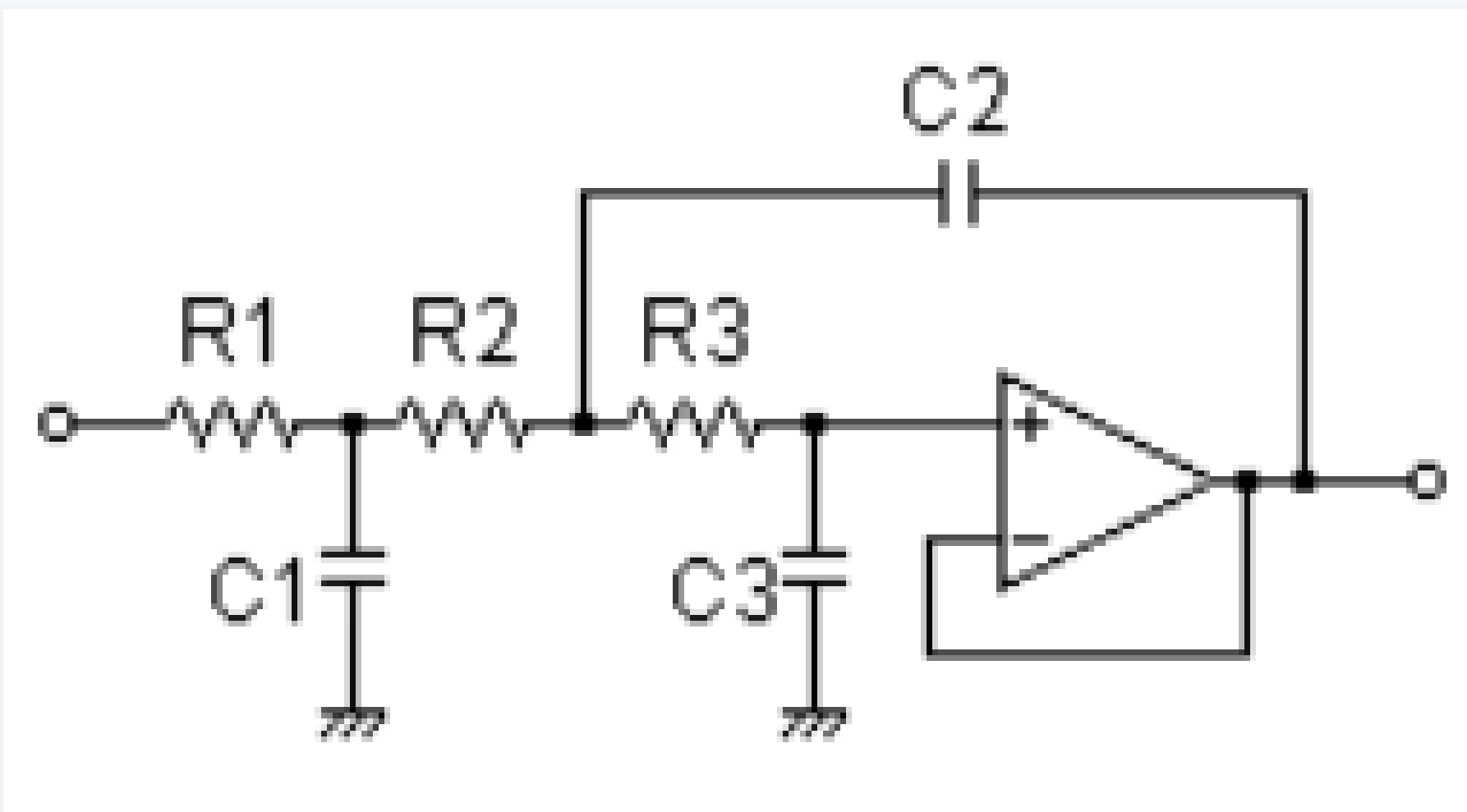


Technical Specifications

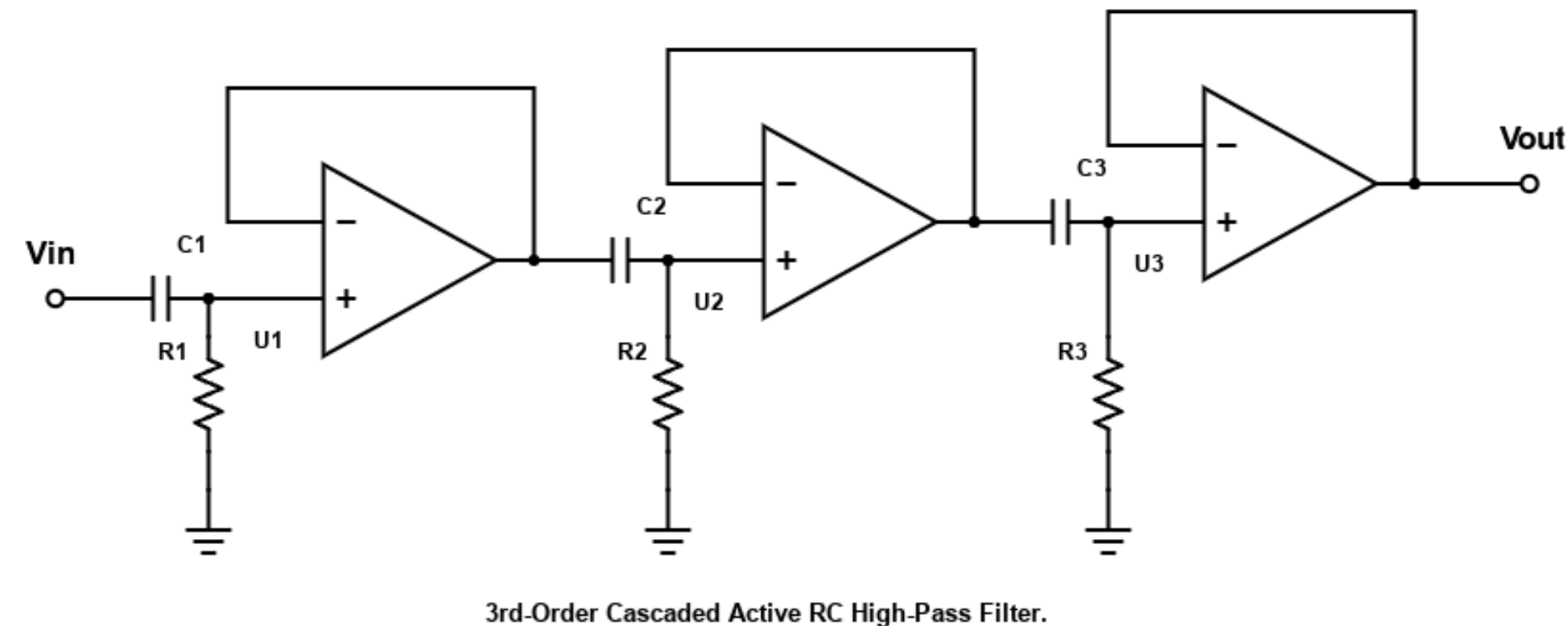
- Filter Order: 3rd-order Active HPF
- Group No.: $G = 4$
- $f_{thres} = (10 + 4) \text{ kHz} = 14 \text{ kHz}$
- Roll-off: -60 dB/decade
- Op-Amp: UA741 (voltage-follower)
- Supply: $\pm 15V$ DC dual-rail

Design 3-Stage Cascaded vs. Sallen-Key

Why cascaded RC + Op-Amp buffer was chosen



1. A single complex circuit with multiple feedback loops.
2. Feedback Sensitive
3. Potential Instability
4. An error in one component affects the entire frequency response.



1. 3 independent 1st-order stages isolated by Op-Amps
2. Active Buffering
3. Better Stability
4. Each stage can be probed individually to find errors

Calculation

3-stage correction factor applied to hit 14 kHz system cutoff

Key Steps

- Target: $f = 14 \text{ kHz}$
- $n = 3$ stages
- Correction factor: $f_{sys} = f_c \times \sqrt[n]{2^{1/n} - 1}$
- Each stage $f_c = 7137.54 \text{ Hz}$
- $C = 1 \text{ nF}$ (fixed for all stages)
- $R \approx 22.3 \text{ k}\Omega$ (calculated)
- Hardware: $R = 22 \text{ k}\Omega$ (lab standard)
- All 3 stages: identical R & C

Calculation Sheet

Given:

- Threshold frequency = $(10 + G_v) \text{ kHz} = 14 \text{ kHz}$
- Number of stages, $n = 3$

For 3-stage filter:

System cutoff frequency:

$$f_{sys} = f_c \times \sqrt[n]{2^{1/n} - 1}$$

Now substitute values:

$$f_{sys} = 14 \times \sqrt[3]{2^{1/3} - 1} = 14 \times \sqrt[3]{1.26 - 1} = 14 \times \sqrt[3]{0.26} = 14 \times 0.50 \\ 7137.54 \text{ Hz}$$

For each stage:

Each stage cutoff frequency:

$$f_c = f_{sys} = 7137.54 \text{ Hz}$$

Using RC formula:

$$f_c = \frac{1}{2\pi RC}$$

Rearranging for R :

$$R = \frac{1}{2\pi f_c C}$$

Given:

- $C = 1 \text{ nF} = 1 \times 10^{-9} \text{ F}$
- $f_c = 7137.54 \text{ Hz}$

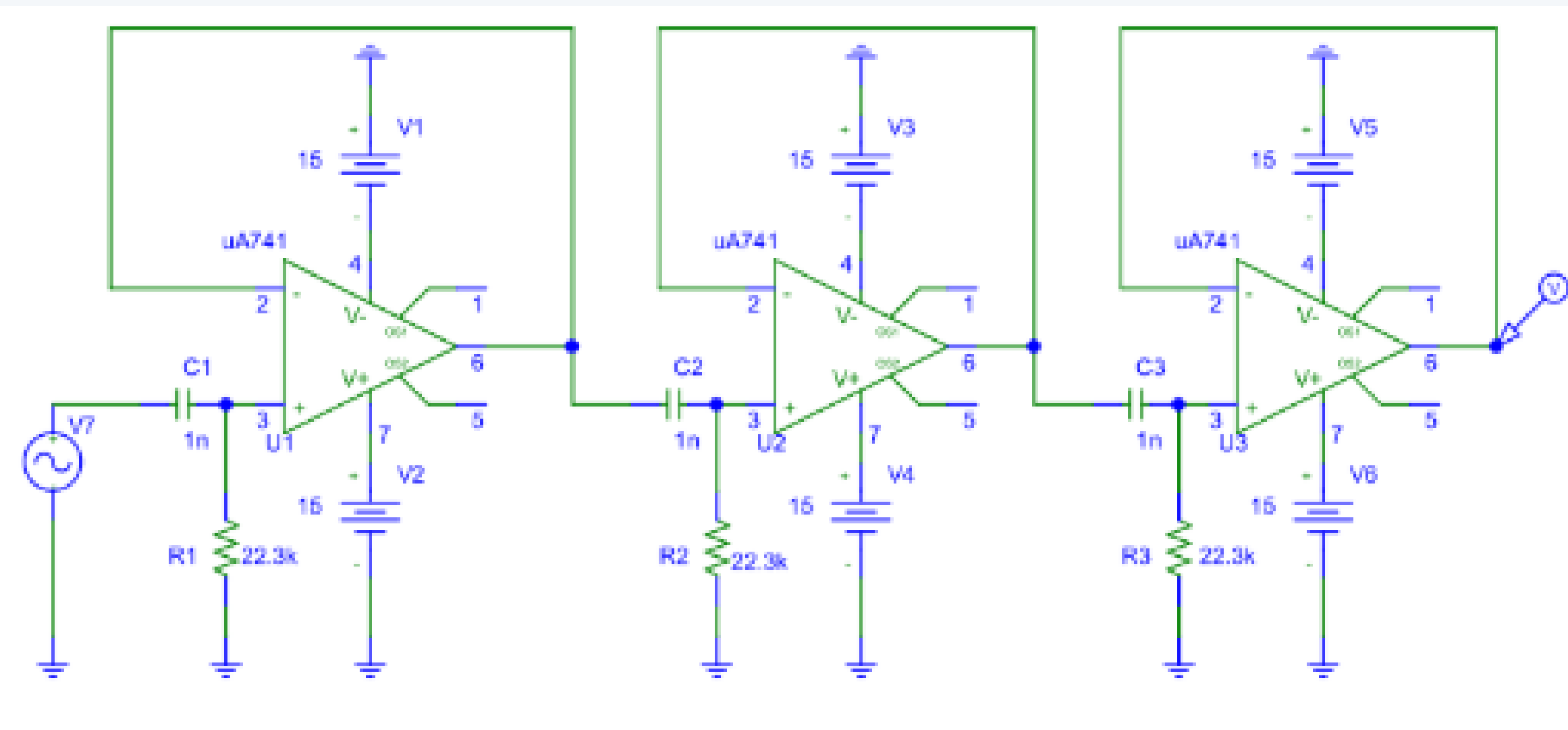
Calculation:

$$R = \frac{1}{2\pi \times 7137.54 \times 10^{-9}} R \approx 22.3 \text{ k}\Omega$$

- Each stage cutoff frequency = **7.137 kHz**
- Component values:
 - $C = 1 \text{ nF}$
 - $R \approx 22.3 \text{ k}\Omega$

Software Implementation — PSpice Circuit Diagram

3-stage cascaded active RC HPF in PSpice schematic

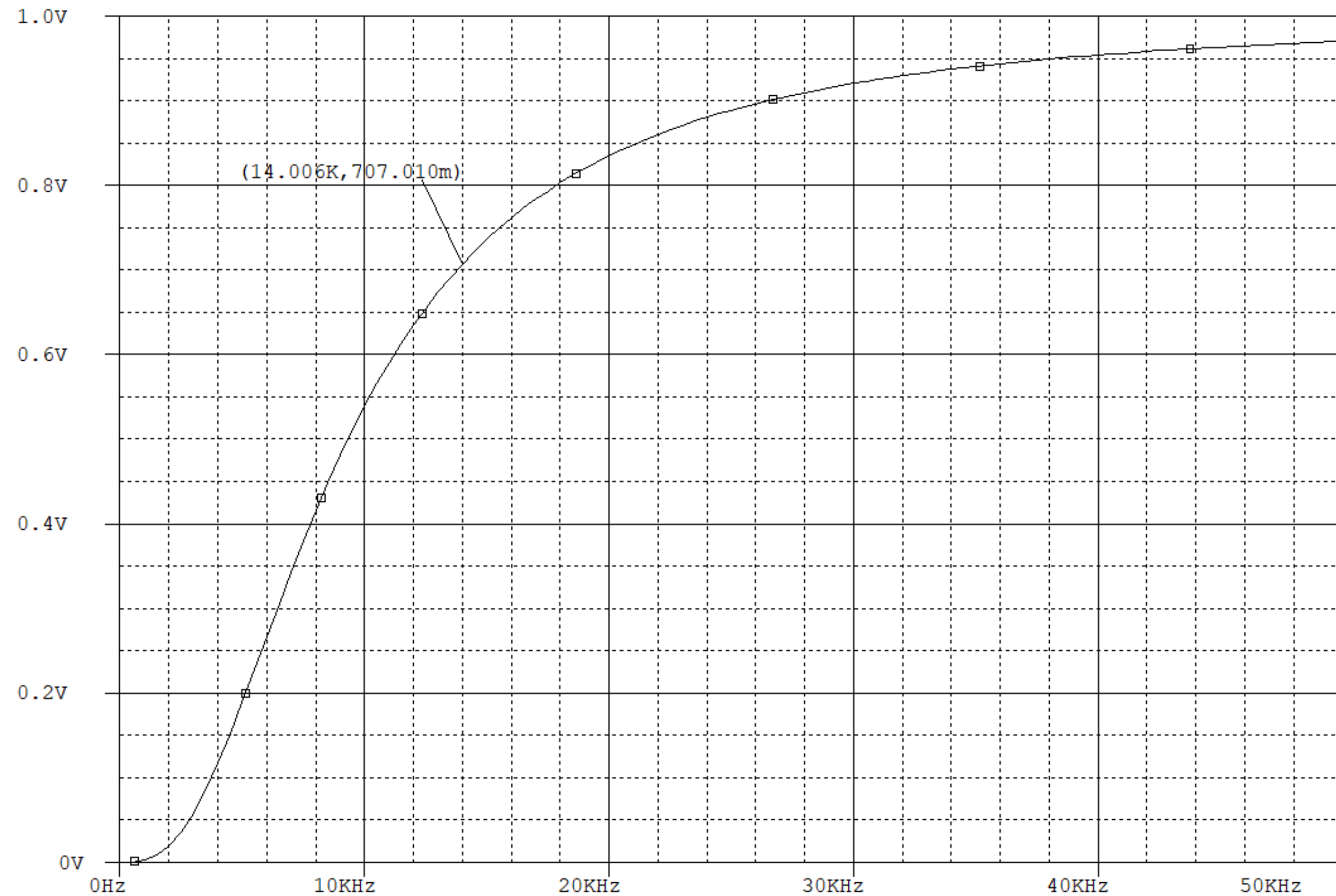


Circuit Notes

- 3 UA741 op-amps
- $R1=R2=R3 = 22.3 \text{ k}\Omega$
- $C1=C2=C3 = 1 \text{ nF}$
- $V+ = +15\text{V}$ (pin 7)
- $V- = -15\text{V}$ (pin 4)
- Vsin input source (V7)
- Voltage marker at Vout
- AC Sweep analysis
- Freq: 0–50 kHz range

Simulation Results — PSpice Frequency Response

AC Sweep output: V(U3:OUT) vs Frequency



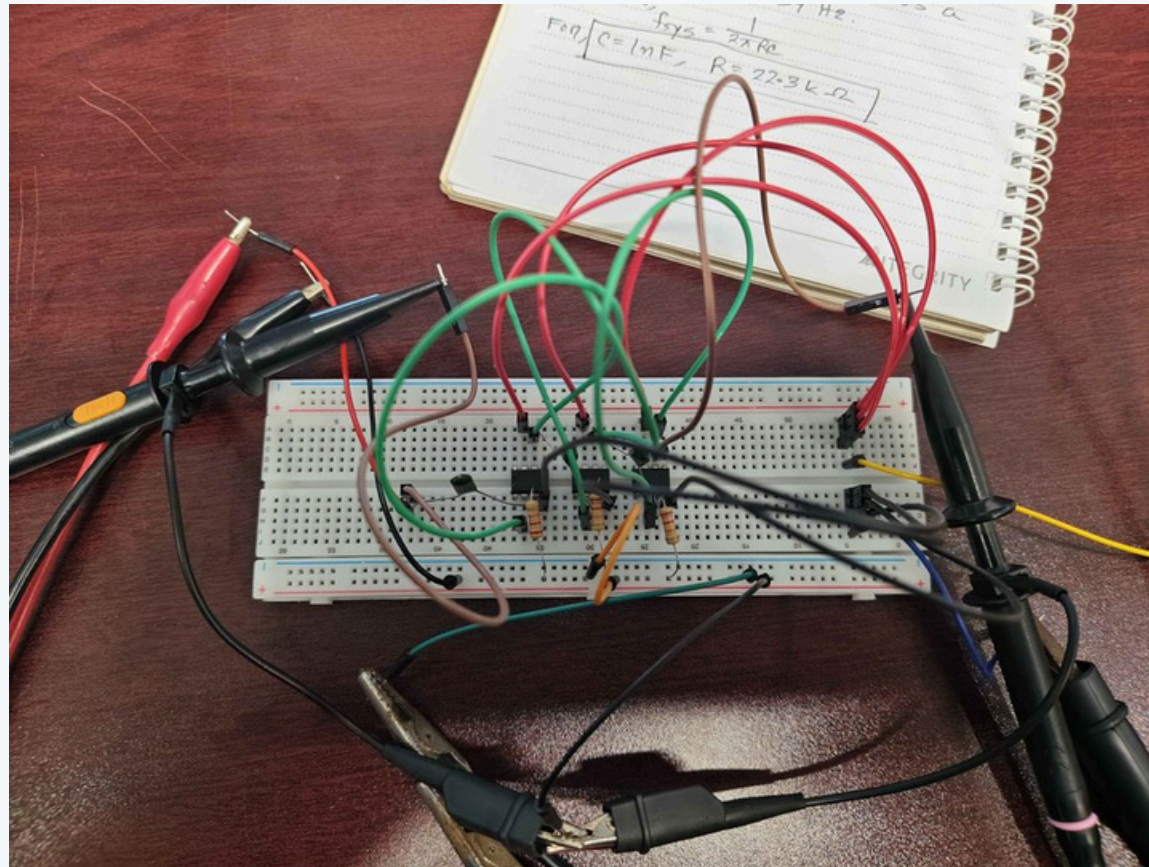
Results Summary

- Cutoff at 0.707 V
- Frequency: 14.006 kHz
- Target: 14.000 kHz
- Error = 0.04%
- Pass-band: flat above 14k
- Stop-band: steep roll-off
- -60 dB/decade confirmed
- Simulation:
SUCCESSFUL

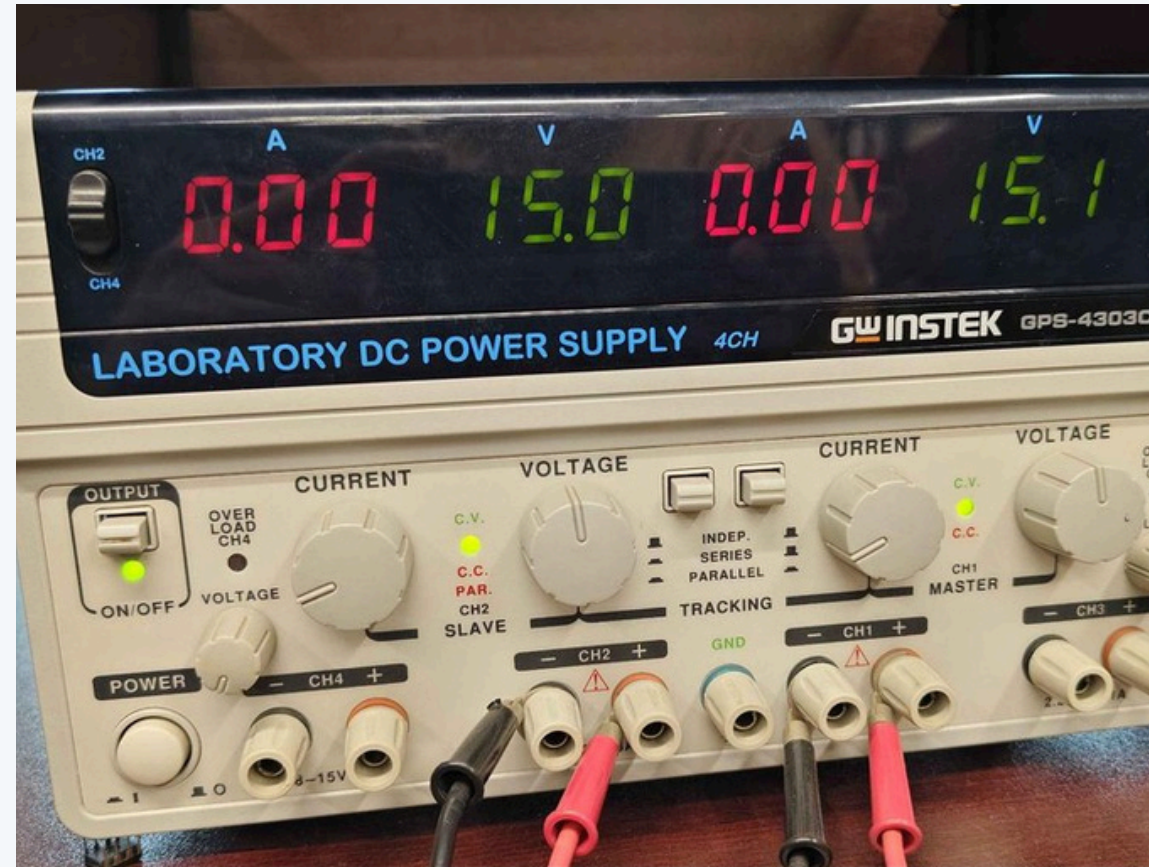
Cutoff point: (14.006K, 707.010m) — shown on graph above

Hardware Implementation — Lab Setup

Breadboard build with UA741, 22 k Ω resistors, 1 nF capacitors



Breadboard Circuit



DC Power Supply $\pm 15V$



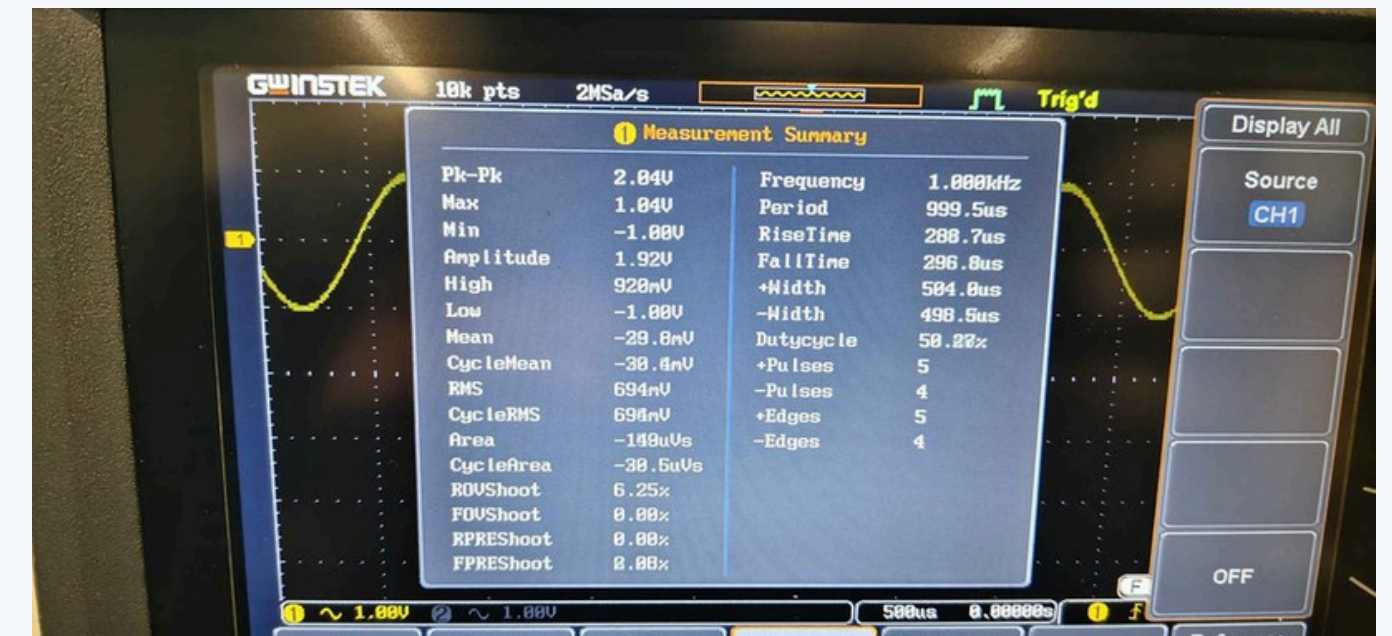
Signal Generator (1kHz start)

Equipment

- Op-Amp: UA741 $\times 3$
- $R = 22\text{ k}\Omega$ ($\times 3$)
- $C = 1\text{ nF}$ ($\times 3$)
- Breadboard
- DC Supply $\pm 15V$
- Oscilloscope GDS-1102B
- Signal Generator SFG-1013
- Jumper wires (color-coded)

Setup Notes

- 22 k Ω used (22.3 unavailable)
- CH1 (Yellow) = Input 1V sine
- CH2 (Blue) = Output
- Probes: 10 $\times$ mode
- Input freq: 1k $\rightarrow$ 40kHz sweep
- +15V $\rightarrow$ Op-Amp Pin 7
- -15V $\rightarrow$ Op-Amp Pin 4



Hardware Results — Stop-Band (Low Frequencies)

Output heavily attenuated below 14 kHz cutoff



f = 1 kHz — Output (CH2 blue line near zero)

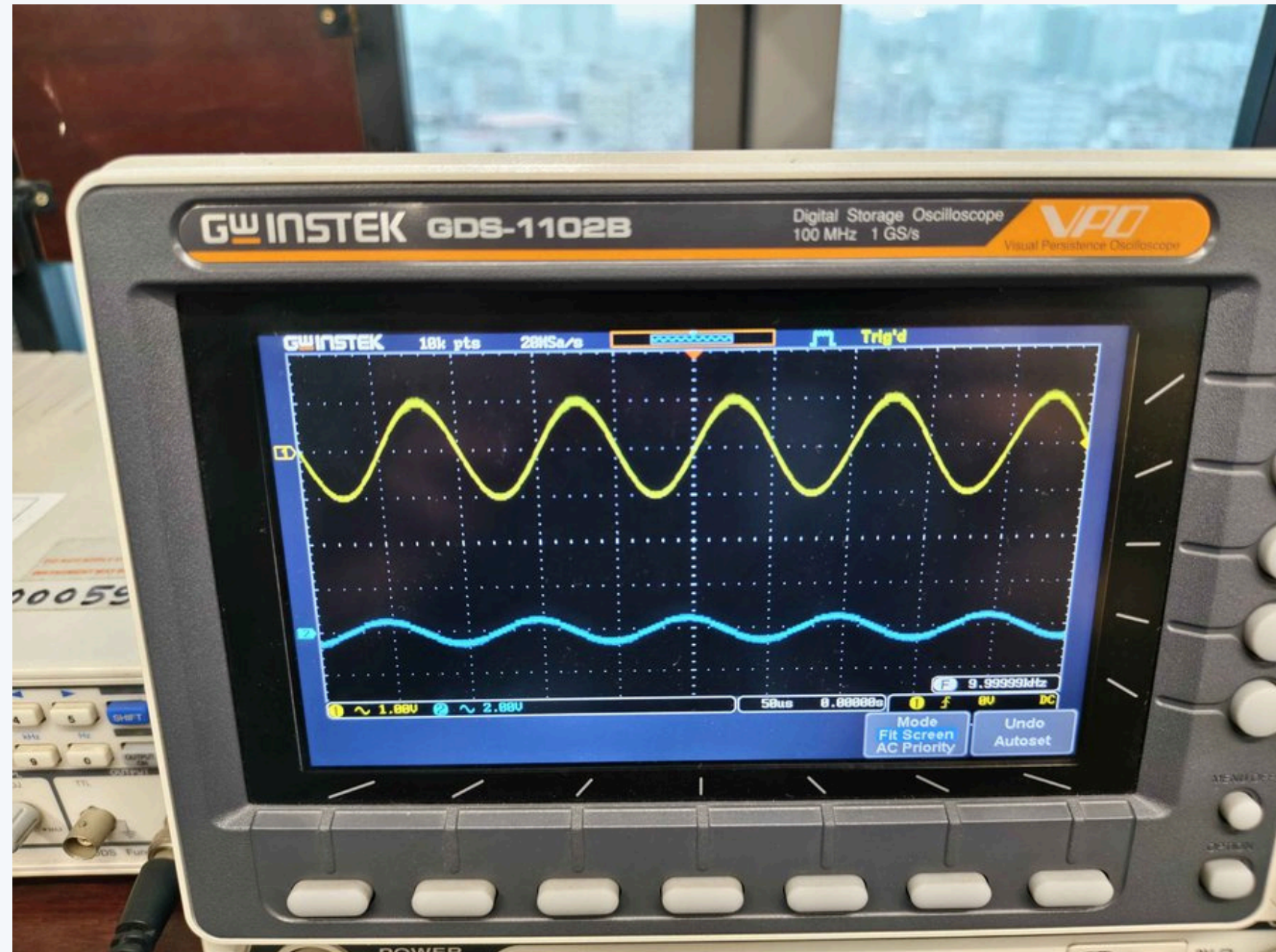
- 1 kHz → output: ~20–50 mV (near zero, flat line on CH2)
- 5 kHz → output: still very low
- Stop-band clearly confirmed



f = 5 kHz — Output still very low, stop-band confirmed

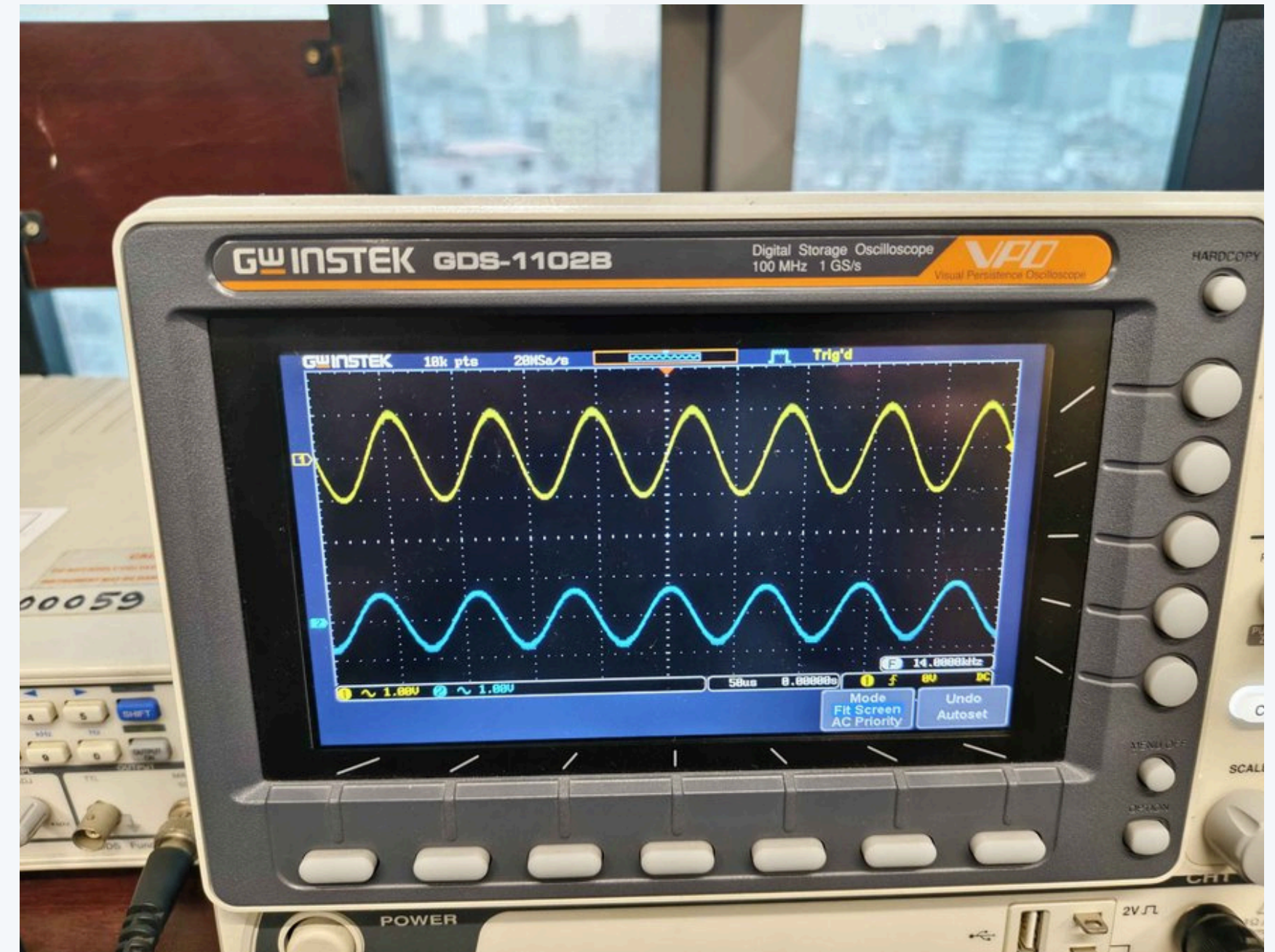
Hardware Results — Transition & Cutoff Frequency

Output rising at 10 kHz, reaching 0.707V threshold at 14 kHz



f = 10 kHz — Output rising (~200 mV), approaching cutoff

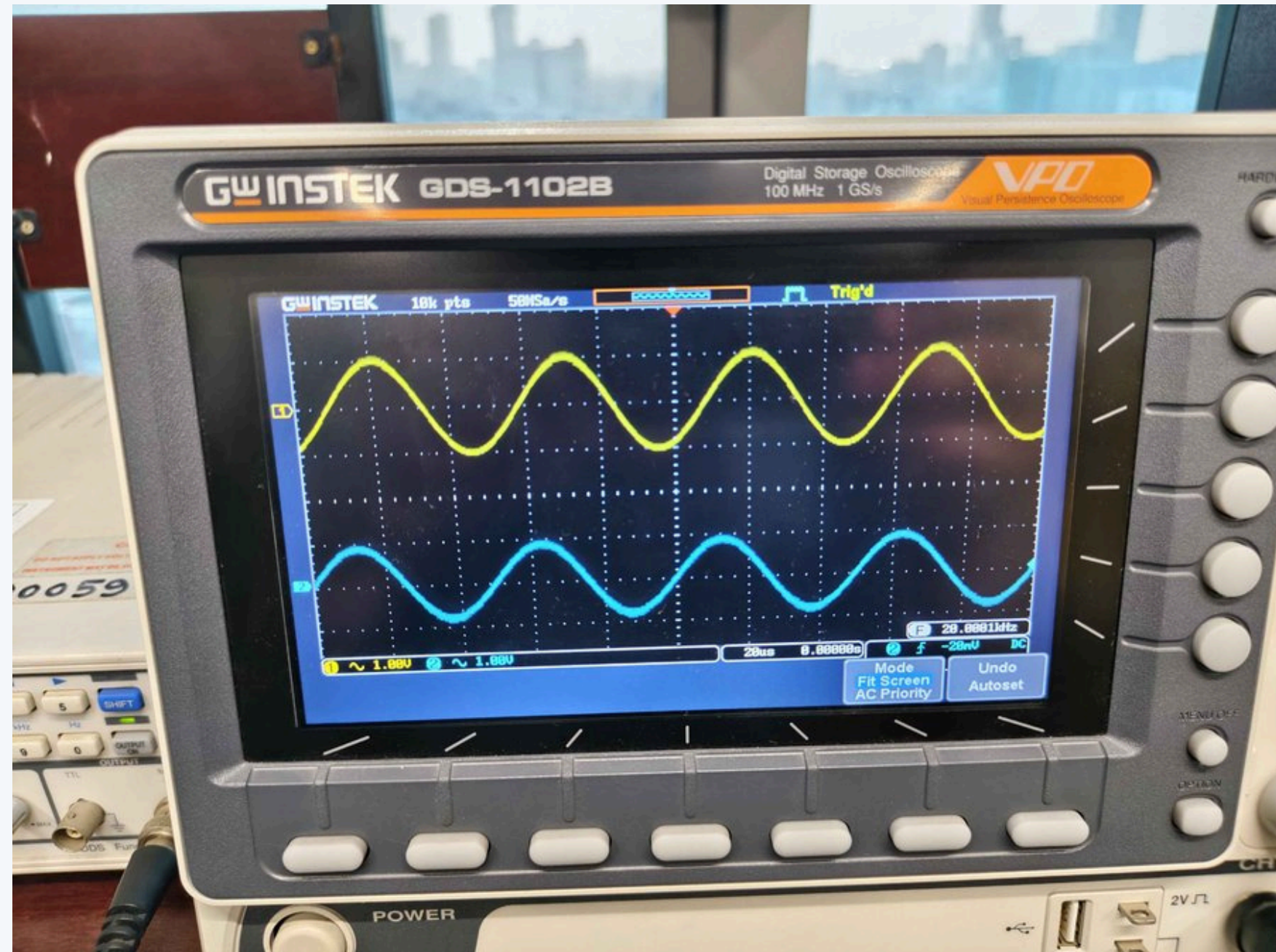
- 10 kHz $\rightarrow$ 200 mV output — transition zone
- 14 kHz $\rightarrow$ 0.720 V measured (expected 0.707 V) — cutoff confirmed



f = 14 kHz — Output $\approx$ 0.720V (target: 0.707V) ✓ Cutoff!

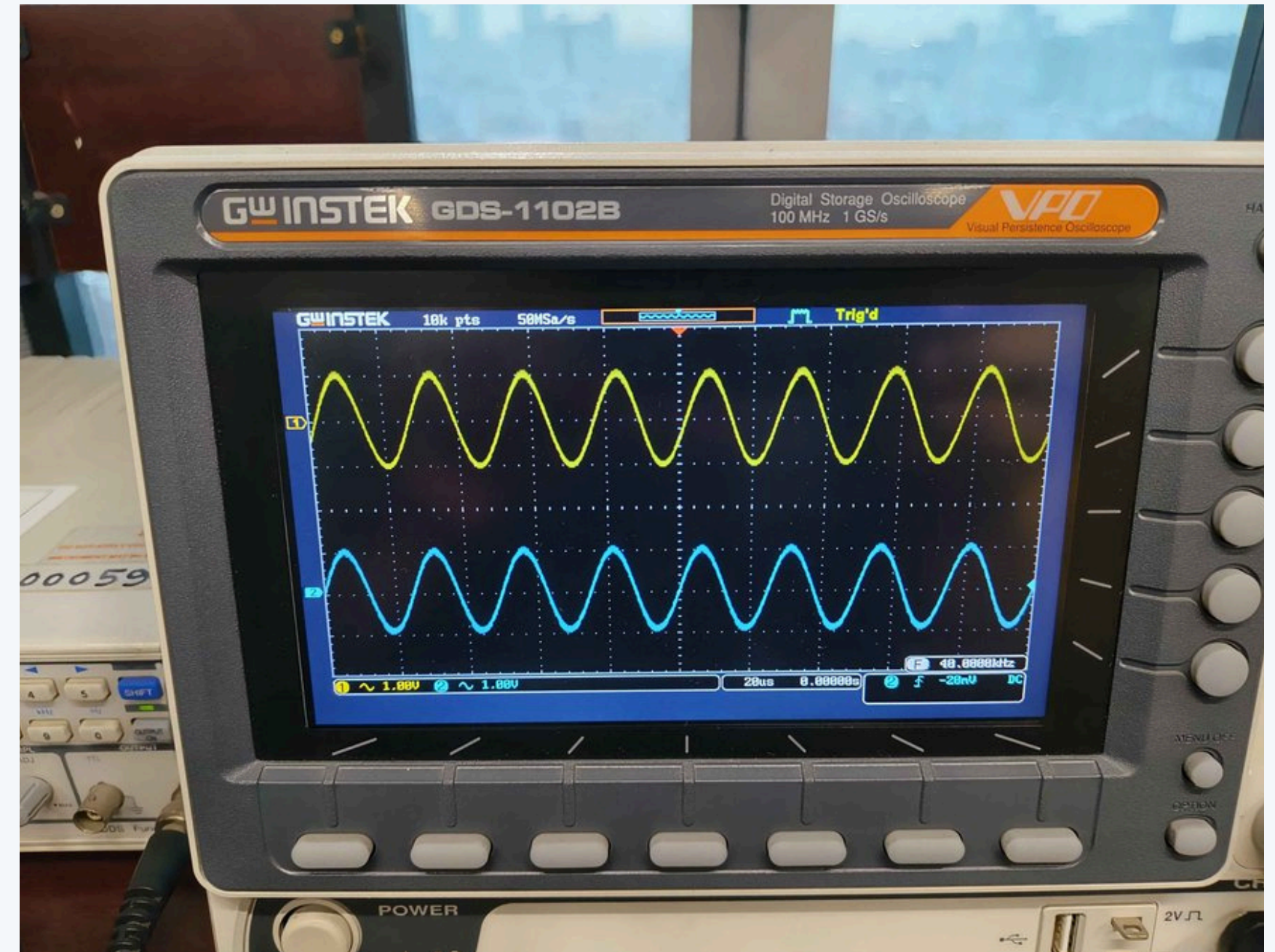
Hardware Results — Pass-Band (High Frequencies)

Filter passes signal with minimal attenuation above 14 kHz



$f = 20 \text{ kHz}$ — Signal passes freely (CH1 & CH2 nearly equal)

- 20 kHz $\rightarrow$ output $\approx$ input, pass-band active
- 40 kHz $\rightarrow$ CH1 & CH2 amplitudes matched — full pass-band confirmed



$f = 40 \text{ kHz}$ — Full pass-band confirmed, both channels matched

Troubleshooting

Issues encountered during hardware lab session

1. Cutoff Frequency Shift

- Simulated: 14.006 kHz | Hardware: ~13.5 kHz
- Cause: 22 k Ω used instead of 22.3 k Ω
- Fix: Accepted as tolerance; future: use potentiometer

2. Op-Amp Malfunction

- No output / distorted signal on power-up
- Cause: Faulty UA741 chips; loose $\pm 15\text{V}$ supply
- Fix: Replaced op-amps; verified dual-rail; stage-by-stage test

3. Noise & Signal Instability

- High-freq noise; jittery waveforms on oscilloscope
- Cause: Long wires as antennas; stray capacitance
- Fix: Shortened wires; color-coded; common ground

4. Loading Effect

- Output amplitude slightly lower than expected
- Cause: 1 $\times$ probe loading; UA741 gain at high freq
- Fix: Switched to 10 $\times$ probes; UA741 linear to 50 kHz

Summary of Findings

Key outcomes from design, simulation & hardware

1. Design Accuracy

- Target fthres = 14 kHz achieved
- Calculations correct & consistent
- 3-stage correction applied properly

2. Simulation Validated

- PSpice cutoff: 14.006 kHz
- Error: 0.04% (< 0.1%)
- Mathematically sound model

3. Hardware Performance

- Low freq (<10kHz): strongly attenuated
- High freq (>14kHz): passes freely
- Output at cutoff: 0.720V (target 0.707V)

4. Practical Challenges

- Faulty UA741 chips replaced
- Component substitution (22k Ω)
- Noise resolved by wire management

5. System Behavior

- Tweeter protected from low-freq damage
- Stable response across all tested ranges
- High-pass function fully verified

6. Design Effectiveness

- Cascaded structure: stable & modular
- Easy per-stage fault detection
- Better than Sallen-Key for lab use

Team Contributions & Conclusion

Member's name	Student ID	Contribution
Mahdi Zaman	24121307	• Circuit Design. • Calculation. • Simulation. • Report Preparation. • Hardware Implementation. • Results Observation. • Error Analysis.
Souvik Barman Ratul	24121205	• Calculation. • Hardware Implementation. • Lab Apparatus Testing. • Report Preparation. • Trial and Error. • Component Selection.
Afifah Bashir Sarah	23221101	• Circuit Design. • Troubleshooting. • Report Preparation. • Hardware Testing.
Sadia Jahin	23221042	• Simulation. • Circuit Design. • Report Preparation.
Fowzia Rahman Orin	23221078	• Troubleshooting. • Hardware Testing. • Report Preparation. • Logistics.

Thank You